ADA Very Important Question CSE/AD/DS/CI/AI-2026

1. Define Algorithm explain steps involved in algorithm design and analysis process with a neat diagram.
2. Explain order of growth with an example order the following to their order growth from the lowest to highest!N-2, logN,N3,2 power N, N cube, N square, NlogN
3. Write an Algorithm to linear search find out time complexity for an algorithm.
4. Apply strasssen's algorithm to multiply the given two matrices .

$$A=\begin{matrix}1 & 2\\ 5 & 6\end{matrix} \quad \text{and} \quad B=\begin{matrix}8 & 7\\ 1 & 2\end{matrix}$$

5. Design an algorithm for quick sort algorithm. Apply quick sort on these elements. Sort array list using using Quick Sort EXAMPLE .
6. Using Selection sort letter word EXAMPLE sort in alphabetical order.
7..MYSTRY(n):

```
S<--0
        for i=0 to n do
            s=s+i*i
         return s
```

i) What does this algorithm compute.
ii) What is the basic operation.
iii) How many times the basic operation executed.
Iv)What is the efficiency class of this algorithm.

7. If t1(n) O(g1(n)) and t2(n) O(g2(n)), then show that t1(n) + t2(n) O(max{g1(n), g2(n)}).
8. Explain the general plan for analyzing the efficiency of a non-recursive algorithm. Suggest a non-recursive algorithm to find maximum element in the list of n numbers. Derive its efficiency
9. Define algorithm. Explain asymptotic notations Big Oh, Big Omega and Big Theta. Notations.
10. Design an insertion sort algorithm and obtain its time complexity. Apply insertion sort on these elements. 25,75,40,10,20,
11.Explain the concept of divide and conquer. Design an algorithm for merge sort and derive its time complexity.
12.).Solve the following Recurrence relation
T(n)=0 if n=0T(n)=1+t(n-1) otherwise.
ii)Solve the following Recurrence relation
X(n)=3x(n-1) for n>1 , x(1)=4
13. Find out time complexity. Determine the number of comparisons researching for the pattern CSE using the text VTUCSECOURCE.
.Sort array list using using Quick Sort
14. 65,70,75,80,85,60,55,50,45.
15. Solve the knapsack problem obtain the optimal solution using Broute force technique. Given problemM=10, n=4 ,w1,w2,w3,w3=7,3,4,5 and Profit P1,p2,p3,p4=42,12,40,25
Consider the Algorithm
16.SECRET(n, a):
Minval<-0

```
for i=0 to n-1 do
      If(a[i]<minval )
        Minval=a[i]
     If(a[i]>maxval )
        Maxval=a[i]
```

iv) What does this algorithm compute.

v) What is the basic operation.

vi) How many times the basic operation executed.

vii) What is the efficiency class of this algorithm.

v)write time complexity with notation